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### A METHOD AND A MIXER FOR COOKING A FOOD PRODUCT THAT CONTAINS STARCH

#### Abstract

A method for cooking a food product by using a mixer, the food product comprising starch and the method comprising: providing liquid and the starch into the mixer; mixing the liquid and the starch to form a mixture; heating the mixture for gelatinizing the starch by injecting a first amount of steam into the mixer until a first set point temperature is reached; heating the mixture for gelatinizing the starch further by feeding a heating medium through a heating jacket of the mixer until a second set point temperature is reached which is higher than the first set point temperature; and during the step of heating by feeding the heating medium through the heating jacket, reducing the injecting of steam to a second amount of steam, wherein the second amount of steam is 0 to 30% of the first amount of steam.

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## **Background/Summary**

### **TECHNICAL FIELD**

[0001] The invention relates to food processing technology. More particularly, it is related to a method and a mixer for cooking a food product that comprises starch.

### **BACKGROUND ART**

[0002] Today mixers are widely used in the field of food processing for dissolving, emulsifying or dispersing powder and liquid ingredients into a liquid medium. A benefit of using mixers, such as high shear mixers, is that it is made possible to achieve a food product that is smooth, homogenous and with consistent quality. A wide range of products can be processed in mixers, such as sauces, mayonnaise and recombined milk.

[0003] A specific use of mixers involves starch processing, also referred to as starch cooking, starch preparation or starch gelatinization. If starch is used as a thickener in a food product, the starch processing is often one of the first steps in a food processing scheme. For instance, in case a low fat mayonnaise is to be produced, one of the first steps in the process may be to process the starch. If this step is performed in the high shear mixer, starch granules may be fed into the mixer and in the mixer, these are combined with water, heated and subjected to shear such that a smooth mixture of starch and water is formed, by swelling the starch so that a more viscous mixture is obtained.

[0004] After the mixture is formed, further ingredients may be added into the mixer such that also further processing steps in preparing the food product is performed in the mixer. Alternatively, after the starch is prepared in the mixer, this may be fed into a tank and transported to another piece of equipment performing the further steps in processing the food product.

[0005] Even though it is known to process starch in mixers, there is room for improvement. More particularly, by being able to more precisely control the starch processing, there will be less undercooked and overcooked starch granules, in turn resulting in less starch being needed for obtaining the desired product properties. Reducing food waste has in turn positive effects on both cost efficiency as well as environmental impact.

### **SUMMARY**

[0006] It is an object of the invention to at least partly overcome one or more of the above-identified limitations of the prior art. In particular, it is an object to provide improved starch processing within the food industry, enabling a smaller amount of starch being required. Another object is to improve energy efficiency when using starch as an ingredient in a food product. Still another object is to provide improved quality of the food product made based on a mixture resulting from the starch processing.

[0007] According to a first aspect, it is provided a method for cooking a food product by using a mixer having a vessel at least partially surrounded by a heating jacket, and a steam injector arranged to inject steam into the vessel. The food product comprises starch and the method comprises: providing liquid into the vessel; providing starch into the vessel; mixing the liquid and

the starch to form a mixture inside the vessel; heating the mixture for gelatinizing the starch by injecting steam at a first steam flow rate into the vessel during a first period of time until a first set point temperature is reached; heating the mixture for gelatinizing the starch further by feeding a heating medium through the heating jacket during a subsequent, second period of time until a second set point temperature is reached which is higher than the first set point temperature; and during the step of heating by feeding the heating medium through the heating jacket, reducing the injecting of steam at to a second steam flow rate, wherein in the second steam flow rate is 0 to 30% of the first steam flow rate.

[0008] Herein, “steam flow rate” should be understood as the mass (or amount) of steam that is injected per unit time. The temperature of the steam is higher than the temperature of the mixture. The steam flow rate is therefore directly related to the rate by which the temperature of the mixture increases, which could be measured e.g. in ° C. per minute. Thus, reducing the injecting of steam to a second steam flow rate that is 0 to 30% of the first steam flow rate means that the rate by which the mixture's temperature increases due to steam injection is lower during the heating by feeding the heating medium through the heating jacket, as compared to when steam is injected at the first steam flow rate. Typically, the heating by feeding the heating medium through the heating jacket may comprise heating the mixture such that the rate by which the temperature of the mixture increases due to both jacket heating and steam heating is 0 to 30% lower compared to the rate by which the temperature of the mixture increased during the heating by injecting steam at the first steam flow rate.

[0009] An advantage with heating the starch by using a two-stage approach is that a high quality product can be provided in an energy efficient manner and with low product losses. By rapidly increasing heat during the first stage by using the steam and thereafter, during the second stage, more slowly increasing the temperature by using the heating jacket, a risk of under-cooking as well as a risk of over-cooking starch granules can be lowered while still ensuring a relatively fast cooking time. As an effect of that less starch granules being non-optimally processed, a better yield can be achieved.

[0010] The first set point temperature may be within a range of 20-75% of the second set point temperature. Alternatively, the first set point temperature may be within a range of 20-65% of the second set point temperature. Alternatively, the first set point temperature may be within a range of 10-64% of the second set point temperature. The first set point temperature may be within a range of 5-45° C. The second set point temperature may be within a range of 50-70° C.

[0011] The first period of time may be within a range of 20-50% of the second period of time. Alternatively, the first period of time may be within a range of 30-50% of the second period of time. The first period of time may be within a range of 2-10 min.

[0012] Alternatively, the first period of time may be within a range of 2-8 min. The second period of time may be within a range of 10-30 min.

[0013] The heating during the first period of time may comprise increasing the temperature of the mixture by a first rate during the first period of time, and the heating during the second period of time may comprise increasing the temperature of the mixture by a second rate during the second period of time, wherein the second rate may be lower than the first rate.

[0014] The second rate may be less than 80%, less than 60% or less than 50% of the first rate. The first rate of increasing the temperature may be within a range of 1-10° C. per minute. The second rate of increasing the temperature may be within a range of 1-5° C. per minute.

[0015] The method may further comprise cooling the mixture by decreasing the temperature of the mixture by a third rate during a third period of time, wherein the third rate of decreasing the temperature may be within a range of 1-8° C. per minute.

[0016] The method may further comprise subjecting the mixture to a first shear rate during the first period of time by injecting steam into the vessel at the first steam flow rate, and subjecting the mixture to a second shear rate during the second period of time by rotating an agitator and/or using

the rotor-stator arrangement located inside the vessel, wherein the second shear rate may be lower than the first shear rate.

[0017] According to a second aspect, it is provided a mixer for cooking a food product, said mixer comprising a vessel at least partially surrounded by a heating jacket, a steam injector arranged to inject steam into the vessel, a processor unit and a memory unit that comprises software instructions which when executed by the processor unit cause the mixer to carry out the method according to the first aspect, including all embodiments thereof.

[0018] The mixer may further comprise an agitator and/or a rotor-stator arrangement inside the vessel, wherein the steam injector may be arranged to subject the starch to a first shear rate, and the agitator and/or the rotor-stator arrangement may be arranged to subject the starch to a second shear rate.

[0019] Still other objectives, features, aspects and advantages of the invention will appear from the following detailed description as well as from the drawings.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0020] Embodiments of the invention will now be described, by way of example, with reference to the accompanying schematic drawings, in which

[0021] FIG. 1 is a cross-sectional view of a mixer.

[0022] FIG. 2 is a graph illustrating temperature and shear versus time in a method for cooking a food product that comprises starch.

[0023] FIG. 3 is a flow chart illustrating a method for cooking a food product that comprises starch.

### DETAILED DESCRIPTION

[0024] With reference to FIG. 1 a mixer **100**, such as a high shear mixer, is generally illustrated. The mixer comprises a vessel **102**, often made of stainless steel, provided with a heating jacket **104**. A liquid, such as water, W can be provided into the vessel **102** via a first inlet **106** and starch STA, e.g. in the form of starch granules or powder, can be provided via a second inlet **108**. Even though illustrated that the first and second inlet **106**, **108** are placed in a side wall of the vessel **102** in FIG. 1, it is also possible to have these inlets placed in a bottom of the vessel **102**. To provide for that the starch STA and the liquid W in the vessel **102** may be heated or cooled, a jacket inlet **110** and a jacket outlet **112** are connected to the heating jacket **104** such that a heating medium HM or a cooling medium CM can be fed through the heating jacket **104**.

[0025] Once the starch STA and the liquid W are combined into a mixture MIX, this may be fed out from the vessel **102** via an outlet **114** in form of a food product FP. The food product FP may be a final food product or may be a food product that may be further processed to thereafter form a final food product that can be consumed by humans. In the latter case the mixture MIX, sometimes referred to as starch phase, may as indicated be transferred to another piece of equipment in which this is combined with other food ingredients and processed into a final food product. In the former case food ingredients may be added into the vessel **102** such that the mixture MIX in combination with the other food ingredients form the final food product inside the mixer **100**.

[0026] Further, to provide for that the starch STA and the liquid W are heated efficiently when being processed into the mixture MIX, steam STE may be injected into the vessel **102** via one or several steam injectors **116a,b** placed in the bottom of the vessel **102**.

[0027] To provide for that the starch STA and the liquid W are heated efficiently via the heating jacket **104** during the processing, an agitator **118** may be provided. The agitator **118** serves the purpose of mixing the starch STA and liquid W thereby resulting in that different parts of mixture MIX are subjected to heat exerted from the heating jacket **104** such that even heat distribution can be provided, but also that the starch STA and liquid W are subjected to a shear. By being subjected

to a shear, properties of starch STA are affected and as an effect, a more efficient mixing can be achieved.

[0028] As illustrated, it is further possible to have a rotor-stator arrangement **120** placed in the bottom of the vessel **102**. This arrangement also serves the purpose of subjecting the starch STA and liquid W to shear, resulting in that the mixing is made efficiently.

[0029] To control the operation of the mixer **100**, a control unit **122** may be provided. Even though not illustrated, the control unit **122** may be communicatively connected to the different parts of the mixer **100** such that the operation of these parts can be made on individual basis, e.g. the control unit **122** can be arranged to control the steam injectors **116a**, **116b**, the flow of the media through the heating jacket **104** and the agitator **118** individually. The control unit **122** may comprise a processor unit **124** and a memory unit **126** that comprises software instructions which when executed by the processor unit **124** cause the mixer **100** to carry out the method described herein. Even though not illustrated, temperature sensors may be provided for continuously reporting a temperature of the mixture MIX to the control unit **122**.

[0030] With reference to FIG. 2, to provide for the starch STA is gelatinized efficiently, that is, providing for that a level of undercooked starch is low and/or that a level of overcooked starch is low, it has been found that a two-stage heating approach is advantageous. In one example, in a first stage performed during a first period of time  $\Delta t_{sub.1}$ , starting in an initial point of time  $t_{sub.0}$  and extending to a first point of time  $t_{sub.1}$ , the steam STE can be injected into the mixture MIX. During this stage heating may be provided via the heating jacket **104** as well. This stage continues until a first set point temperature  $T_{sub.1}$  is reached, i.e. until a temperature of the mixture MIX is increased from an initial temperature  $T_{sub.0}$  to the first set point temperature  $T_{sub.1}$ . An effect of injecting steam into the mixture MIX is that rapid heating can be achieved. As illustrated, the temperature of the vessel **102** can be increased at a first rate  $\alpha_{sub.1}$ . The first rate  $\alpha_{sub.1}$  is the temperature increase per unit time, which could be measured as ° C. per minute. During this first stage, steam is injected at a first steam flow rate.

[0031] Once having reached the first set point temperature  $T_{sub.1}$  and until a second set point temperature  $T_{sub.2}$  is reached, during a second period of time  $\Delta t_{sub.2}$ , running from the first point of time  $t_{sub.1}$  to a second point of time  $t_{sub.2}$ , the temperature of the mixture MIX is increased by that the mixture MIX is heated via the heating jacket **104** either in isolation or in combination with that the steam STE is injected via the steam injector **116a,b** at a much lower steam flow rate. Thus, during this second period of time  $\Delta t_{sub.2}$ , the steam STE is injected at a second steam flow rate that is lower than the first steam flow rate, such as 0 to 30% of the first steam flow rate. This means that the steam injection may, in one embodiment, be fully stopped during the second period of time  $\Delta t_{sub.2}$ . As an effect of lowering or cutting off steam inflow into the vessel **102**, a second rate  $\alpha_{sub.2}$ , representing temperature increase rate during the second period of time  $\Delta t_{sub.2}$ , is lower than the first rate  $\alpha_{sub.1}$ .

[0032] Each of the first and second set point temperatures  $T_{sub.1}$ ,  $T_{sub.2}$  may be reached by using both steam and the heating jacket during both periods of time  $\Delta t_{sub.1}$ ,  $\Delta t_{sub.2}$ . However, the second rate  $\alpha_{sub.2}$  is lower than the first rate  $\alpha_{sub.1}$ . This is preferably accomplished by injecting steam STE at the second steam flow rate during the second period of time  $\Delta t_{sub.2}$ , with the second steam flow rate being 0 to 30% of the first steam flow rate which was used during the first period of time  $\Delta t_{sub.1}$ .

[0033] By using first quicker heating, i.e. a first and higher heating rate  $\alpha_{sub.1}$ , and thereafter a second slower heating, i.e. a second and lower heating rate  $\alpha_{sub.2}$ , gelatinizing the starch STA can be improved both in terms of quality, i.e. reduced number of undercooked and/or overcooked starch granules, as well as energy efficiency.

[0034] During a third period of time  $\Delta t_{sub.3}$ , running from the second point of time  $t_{sub.2}$  to a third point of time  $t_{sub.3}$ , the temperature can be kept at the second set point temperature  $T_{sub.2}$ . During this period of time, shear and agitation may be provided via the agitator **118** and/or the

rotor-stator arrangement **120**.

[0035] During a fourth period of time  $\Delta t_{\text{sub.4}}$ , running from the third point of time  $t_{\text{sub.3}}$  to a fourth point of time  $t_{\text{sub.4}}$ , the temperature can be lowered at a third rate  $\alpha_{\text{sub.3}}$  from the second set point temperature  $T_{\text{sub.2}}$  to a third set point temperature  $T_{\text{sub.3}}$ . In this cooling phase, the cooling medium CM may be fed into the heating jacket **104**. Even though not illustrated, it is also possible to cool down the mixture MIX by using an external device, such as a scraped surface heat exchanger (SSHE), a plate heat exchanger (PHE), or a tubular heat exchanger (THE). The mixture MIX may be fed to this external device and then fed back to the mixer **100** for further processing or the mixture MIX may be fed downstream to other pieces of equipment for further processing. An advantage of having the external device for cooling down the mixture MIX is that a risk of overheating the mixture MIX can be reduced, thereby reducing the risk of overcooking the starch STA, which may in the form of a starch phase or in a starch-thickened condiment sauce, e.g. marinade or barbecue sauce. By being able to reduce the amount of overcooked starch, i.e. by improving the yield of the starch, it is made possible to achieve a high viscous food product from a low amount of starch. In addition to delivering high quality product properties, the improved viscosity also has an effect of shelf life, i.e. the properties of the food product can be maintained for a longer period of time.

[0036] As illustrated in FIG. 2, the shear subjected to the mixture MIX is different during different periods of time. In the example illustrated, a first shear rate  $\beta_{\text{sub.1}}$ , illustrated by dotted line, can be provided during the first period of time  $\Delta t_{\text{sub.1}}$  and a second shear rate  $\beta_{\text{sub.2}}$ , being less than the first shear rate **1**, can be provided during the second, third and fourth period of time  $\Delta t_{\text{sub.2}}$ ,  $\Delta t_{\text{sub.3}}$ ,  $\Delta t_{\text{sub.4}}$ . The first shear rate  $\beta_{\text{sub.1}}$  may be at least 50% greater than the second shear rate  $\beta_{\text{sub.2}}$ . By adjusting the flow rate of steam STE injected during the second period of time  $\Delta t_{\text{sub.2}}$ , the shear rate can be affected. Further, the shear rate may also be affected by the rotor-stator arrangement **120** and/or the agitator **118**. During the fourth period of time  $\Delta t_{\text{sub.4}}$ , the shear rate may also be affected by the external device used for cooling the mixture MIX.

[0037] Even though the temperatures and periods of time may be different for different mixtures MIX, and in turn different food products, it has been found that the first set point temperature  $T_{\text{sub.1}}$  may be within a range of 20-75% of the second set point temperature  $T_{\text{sub.2}}$ . Further, it has been found that the first set point temperature  $T_{\text{sub.1}}$  can be within a range of 5-45° C. The second set point temperature  $T_{\text{sub.2}}$  may be within a range of 50-70° C. Regarding the periods of times being used, the first period of time  $\Delta t_{\text{sub.1}}$  may be within a range of 20-50% of the second period of time  $\Delta t_{\text{sub.2}}$ . The first period of time  $\Delta t_{\text{sub.1}}$  may be within a range of 2-10 min. The second period of time may be within a range of 10-30 min. Regarding the first, second and third rates  $\alpha_{\text{sub.1}}$ ,  $\alpha_{\text{sub.2}}$ ,  $\alpha_{\text{sub.3}}$ , these may also vary for different food products, but it has been found beneficial to have the second rate  $\alpha_{\text{sub.2}}$  is 80% of the first rate  $\alpha_{\text{sub.1}}$  or lower. The first rate  $\alpha_{\text{sub.1}}$  of increasing the temperature may be within a range of 1-10° C. per minute. The second rate of increasing the temperature may be within a range of 1-5° C. per minute. These values applies for most starch types commonly used in food products.

[0038] In practical applications the rates  $\alpha_{\text{sub.1}}$ ,  $\alpha_{\text{sub.2}}$ ,  $\alpha_{\text{sub.3}}$  may not always be as linear as illustrated in FIG. 2. Instead they may look more like one of the lines illustrated by dashed lines I and II. By being able to control the heating, the shear rate and also the cooling as described above, it is made possible to achieve a high yield during starch processing, which may be 10% or less under-cooked starch granules, at least 80% properly cooked starch granules, and 10% or less over-cooked starch granules.

[0039] FIG. 3 is a flowchart illustrating a method **200** for cooking a food product which includes gelatinizing starch STA by using the mixer **100** illustrated in FIG. 1. As illustrated, in a first step **202**, the liquid W can be provided into the vessel **102** via the first inlet **106**. In parallel, before or after, in a second step **204**, the starch STA can be provided via the second inlet **108**. After having received the liquid W and the starch STA, in a third step **206**, the liquid W and the starch STA can

be mixed into the mixture MIX. In a fourth step **208**, the mixture MIX can be heated by the steam STE injected into the vessel **102**. After being heated by the steam STE, in a fifth step **210**, the mixture MIX can be heated by using the heating jacket **104**. As described above, the steam STE may also be injected during the fifth step **210**, but at a lower steam flow rate compared to the fourth step **208**.

[0040] In line with the graph presented in FIG. **2**, the fifth step **210** may be followed by a sixth step **212** in which the mixture MIX is cooled. The cooling may be provided by that the cooling medium is fed through the heating jacket **104**, but it is also possible, as described above, to cool down the mixture MIX via the external device.

[0041] In parallel with the fourth step **208**, the mixture MIX may be subjected to the first shear rate  $\beta_{\text{sub.1}}$  in a seventh step **214**. This first shear rate  $\beta_{\text{sub.1}}$  may be formed by the steam STE injected via the steam injector **116a,b**.

[0042] In parallel with the fifth step **210**, while the mixture MIX is heated via the heating medium HM fed through the heating jacket **104**, the mixture MIX may be subjected to a second shear rate  $\beta_{\text{sub.2}}$  in an eighth step **216**. The shear in this step may be generated by the agitator **118** and/or the rotor-stator arrangement **120**. The shear may also be generated, in combination with the agitator **118** and/or the rotor-stator arrangement **120**, by the steam STE injected via the steam injectors **116a, b**, but the second steam flow rate used during this eighth step is lower than the first steam flow rate used during the seventh step **214**.

[0043] From the description above follows that, although various embodiments of the invention have been described and shown, the invention is not restricted thereto, but may also be embodied in other ways within the scope of the subject-matter defined in the following claims.

## Claims

1. A method for cooking a food product by using a mixer having a vessel at least partially surrounded by a heating jacket, and a steam injector arranged to inject steam into the vessel, the food product comprising starch and the method comprising providing liquid into the vessel, providing the starch into the vessel, mixing the liquid and the starch to form a mixture inside the vessel, heating the mixture for gelatinizing the starch by injecting steam at a first steam flow rate into the vessel during a first period of time until a first set point temperature is reached, heating the mixture for gelatinizing the starch further by feeding a heating medium through the heating jacket during a subsequent, second period of time until a second set point temperature is reached which is higher than the first set point temperature, and during the step of heating by feeding the heating medium through the heating jacket, reducing the injecting of steam to a second steam flow rate, wherein the second steam flow rate is 0 to 30% of the first steam flow rate.
2. The method according to claim 1, wherein the first set point temperature is within a range of 20-75% of the second set point temperature.
3. The method according to claim 1, wherein the first set point temperature is within a range of 5-45° C.
4. The method according to claim 1, wherein the second set point temperature is within a range of 50-70° C.
5. The method according to claim 1, wherein the first period of time is within a range of 20-50% of the second period of time.
6. The method according to claim 1, wherein the first period of time is within a range of 2-10 min.
7. The method according to claim 1, wherein the second period of time is within a range of 10-30 min.
8. The method according to claim 1, wherein the heating during the first period of time comprises increasing the temperature of the mixture by a first rate during the first period of time, and the heating during the second period of time comprises increasing the temperature of the mixture by a

second rate during the second period of time, the second rate being lower than the first rate.

**9.** The method according to claim 8, wherein the second rate is less than 80% of the first rate, or lower.

**10.** The method according to claim 8, wherein the first rate of increasing the temperature is within a range of 1-10° C. per minute.

**11.** The method according to claim 8, wherein the second rate of increasing the temperature is within a range of 1-5° C. per minute.

**12.** The method according to claim 1, comprising cooling the mixture by decreasing the temperature of the mixture by a third rate during a third period of time, wherein the third rate of decreasing the temperature is within a range of 1-8° C. per minute.

**13.** The method according to claim 1, comprising subjecting the mixture to a first shear rate during the first period of time by injecting steam into the vessel at the first steam flow rate, and subjecting the mixture to a second shear rate during the second period of time by rotating an agitator and/or using the rotor-stator arrangement located inside the vessel, wherein the second shear rate is lower than the first shear rate.

**14.** A mixer for cooking a food product, said mixer comprising a vessel at least partially surrounded by a heating jacket, a steam injector arranged to inject steam into the vessel, a processor unit and a memory unit that comprises software instructions which when executed by the processor unit cause the mixer to carry out the method according to claim 1.

**15.** The mixer according to claim 14, comprising an agitator and/or a rotor-stator arrangement inside the vessel, wherein the steam injector is arranged to subject the starch to a first shear rate, and the agitator and/or the rotor-stator arrangement are arranged to subject the starch to a second shear rate.

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